PROJECT OVERVIEW & ILLUSTRATED USER GUIDE

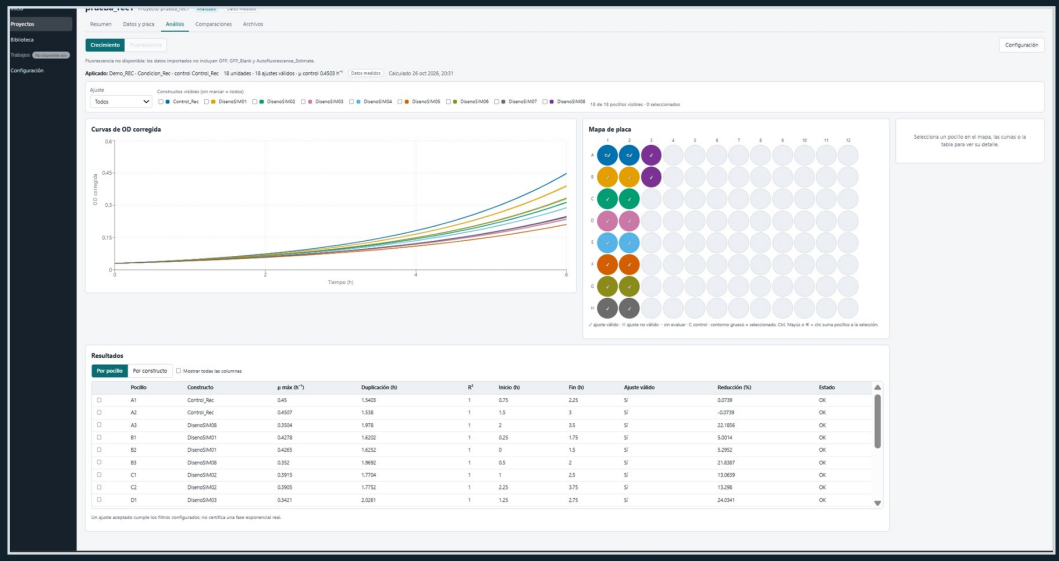
SynBio Learn

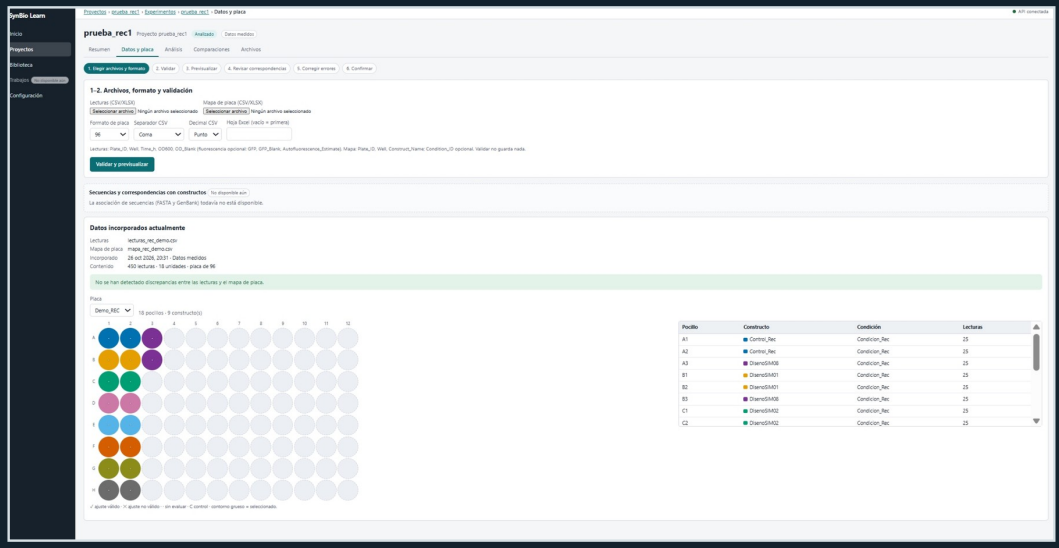
Microplate analysis and experiment management for synthetic biology

A personal scientific software project developed in Python. It links readings, plate layouts and constructs, analyses growth and fluorescence, and keeps results ready to compare and export.

Prototype · Development version

October 2026





Source code maintained privately. Demonstration and selected results available on request.

Contents and how to read this guide

OVERVIEW

At a glance 3

Workflow and application map 4

FEATURE GUIDE

Experiment overview 5

Data import and plate layout 7

Growth analysis 9

Fluorescence analysis 11

Saved comparisons 13

Files and exports 15

Construct library 17

REFERENCE AND ROADMAP

Glossary 19

Development roadmap 20

Jobs module: concept and example 21

Jobs module: design goals 22

Quality and availability 23

Status badges

IN PROTOTYPE

Present in the current prototype and shown in the screenshots.

IN PROGRESS

Being integrated or developed; not yet a finished capability.

PLANNED

Designed, but not yet developed.

About the examples

The screenshots come from example experiments used to illustrate the interface, such as prueba_rec1 and fluorescencia1, with constructs named Demo or SIM. They show how information is organised and are not presented as experimental findings.

The application interface is in Spanish. Spanish labels appear in parentheses next to the English terms used here.

What SynBio Learn does

Microplate experiments produce many readings that only make sense when each one is tied to a well, a construct and an experimental condition. SynBio Learn keeps those links explicit. Readings and plate maps are imported together, checked against each other, analysed and stored with the experiment, so results can be revisited, compared and exported.

1 Organise

Projects, experiments, descriptive context and a library of constructs, kept together.

2 Import and validate

CSV or Excel readings and plate maps, checked against each other and previewed before they are confirmed.

3 Analyse

Growth curves with growth-rate and doubling-time estimates, fit windows and quality indicators, plus OD-normalised fluorescence.

4 Compare and export

Saved analyses side by side by construct, with original inputs and CSV outputs inside the application.

Capabilities and status

AREA	CAPABILITY	STATUS
Experiment management	Projects, experiments, context fields, construct library	IN PROTOTYPE
Data import	CSV/Excel readings and plate maps, validation, plate preview	IN PROTOTYPE
Growth analysis	Corrected-OD curves, μ, doubling time, R^2, fit window, reduction vs control	IN PROTOTYPE
Fluorescence	OD-normalised fluorescence with validity counts (needs fluorescence columns)	IN PROTOTYPE
Comparison and export	Saved-analysis comparison, CSV outputs, access to original files	IN PROTOTYPE
Sequence viewer	Inspect sequences linked to constructs	IN PROGRESS
Machine learning	Learning from accumulated results; recommender ranking interface	IN DEVELOPMENT
Biophysical models	Model-based analysis layer	IN DEVELOPMENT
Jobs	Background monitoring of long calculations	PLANNED

Workflow and application map

1 Organise	2 Import	3 Validate	4 Analyse	5 Compare	6 Export
Create a project and an experiment and record its context.	Load readings and a plate map in CSV or Excel.	Check, preview and correct matches before confirming.	Growth curves and OD-normalised fluorescence.	Place saved analyses side by side by construct.	Open the original files and CSV outputs.

Main navigation (side menu)

Home (Inicio)
Recent experiments with their status, an area for notices, and shortcuts to create or open a project and to consult the library.

Projects (Proyectos)
Projects and the experiments they contain.

Library (Biblioteca)
Constructs recorded across projects. A Sequences tab is reserved for the sequence module.

Jobs (Trabajos)
Planned monitor for long calculations; shown as not yet available.

Settings (Configuración)
Application settings.

Inside an experiment (tabs)

Summary (Resumen)
Experimental context and experiment status.

Data and plate (Datos y placa)
Import, validation and plate view.

Analysis (Análisis)
Growth and fluorescence results.

Comparisons (Comparaciones)
Saved analyses side by side.

Files (Archivos)
Original files and exports.

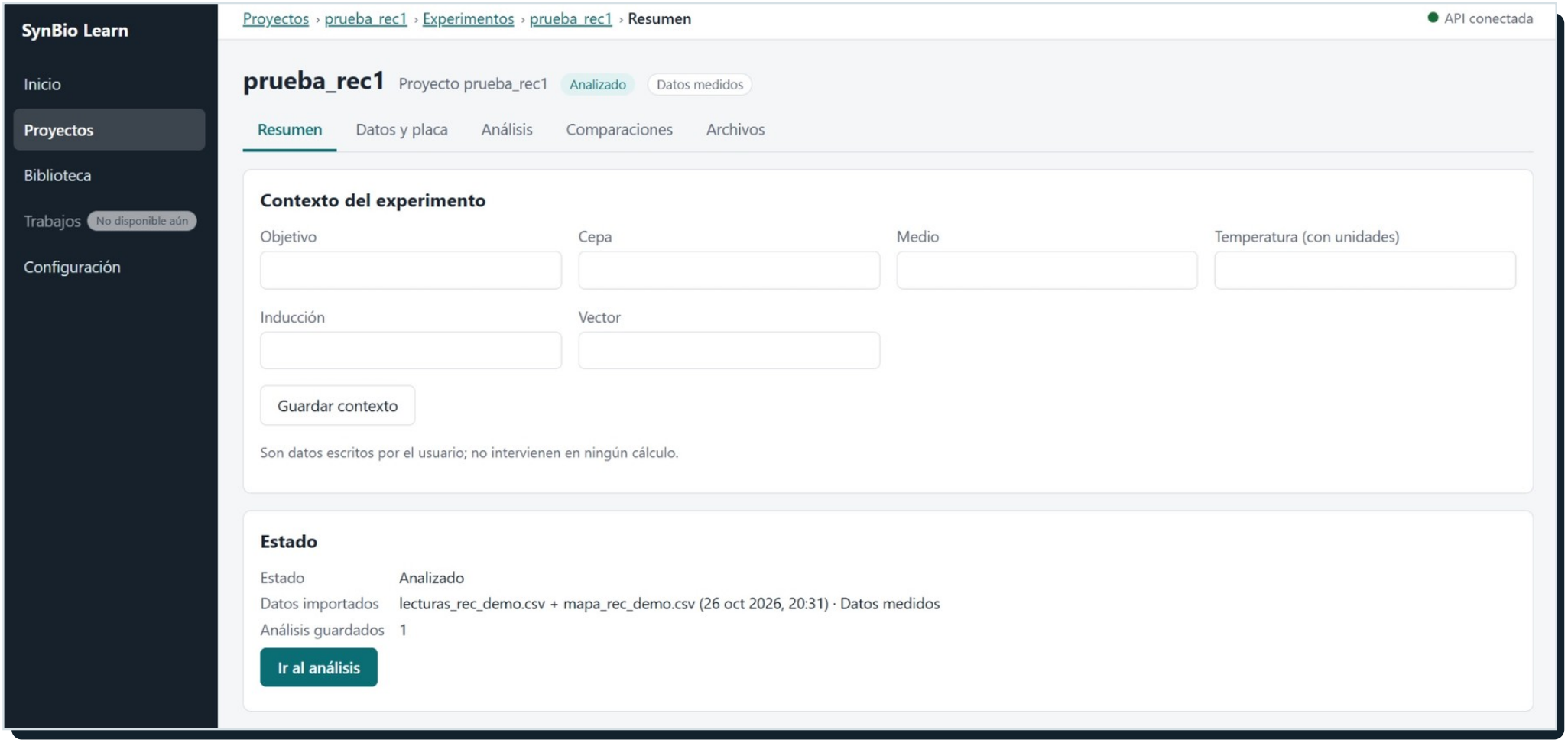
A status indicator at the top right of each screen shows whether the application's API is connected ("API conectada").

One colour per construct
Each construct keeps the same colour in the plate map, the curves and the construct filters.

Badges on every tab
State and data-origin badges accompany the experiment name throughout.

Select from anywhere
A well can be selected from the plate map, the curves or the table to see its detail.

Experiment overview



What you see

Experiment context form, a status panel with imported files and saved analyses, and state badges beside the experiment name.

Main controls

Context fields with Save context, the tab bar, and the Go to analysis button.

What it gives you

Context stored with the experiment and a quick check of its data and analysis history.

Experiment overview: how to use it

How to use this view

1. Open the experiment from Projects, or from the recent-experiment list on the home screen.
2. Fill in the context: objective, strain, medium, temperature (with units), induction and vector. Select Save context.
3. Check the Status panel for the state, the imported files with their import date and the number of saved analyses.
4. Move on with the tab bar, or select Go to analysis to open the results.

What the information means

Breadcrumb trail
Projects › project › Experiments › experiment › Summary shows where you are and lets you step back to any level.

State badge
Shows how far the experiment has progressed, for example Analysed (Analizado).

Data-origin badge
Labels the data, for example Measured data (Datos medidos) or Demonstration (Demostración), so that example and real datasets stay distinguishable.

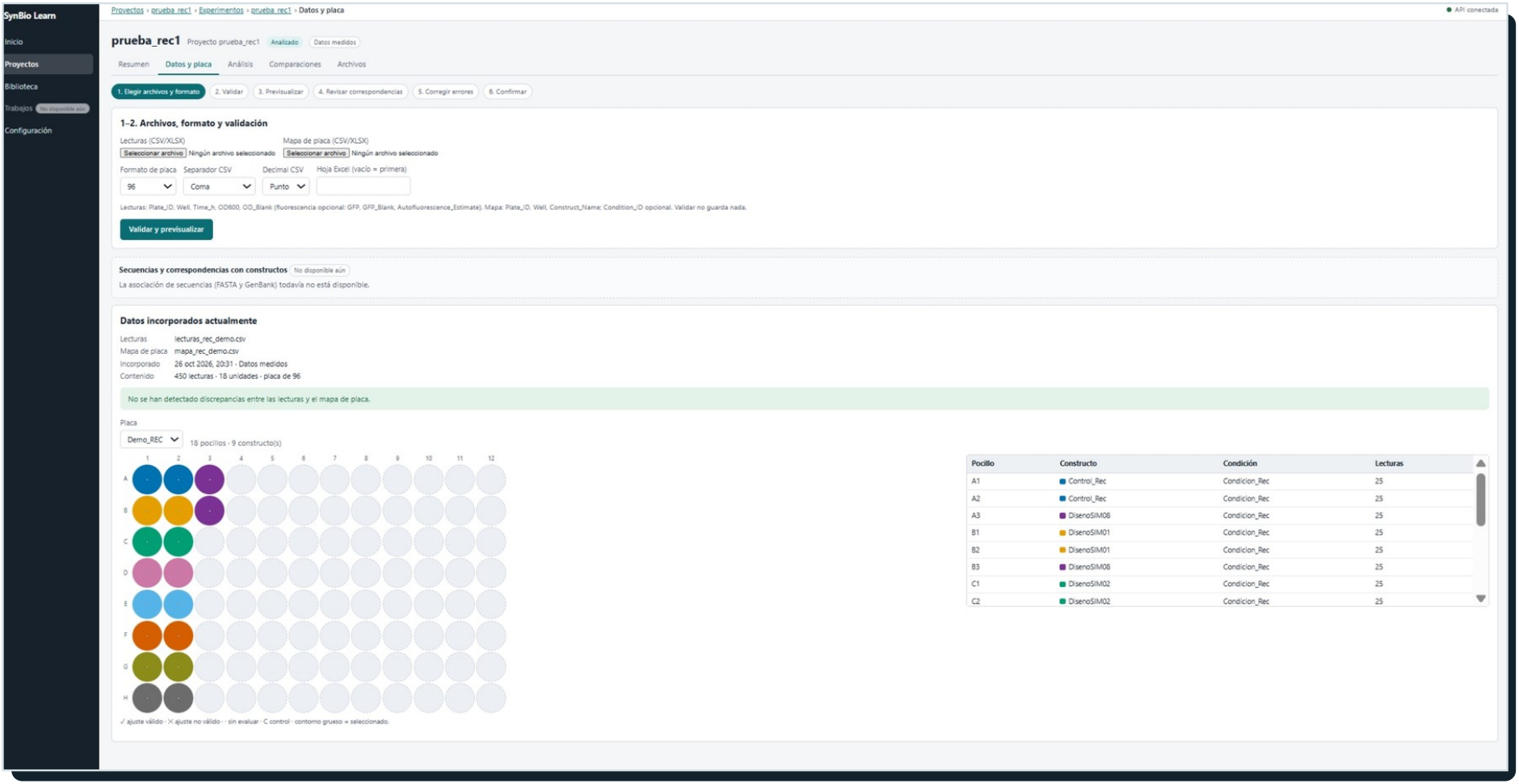
Imported data
Names the readings file and the plate-map file, with the import date and the data-origin label.

Saved analyses
Counts the analyses stored for the experiment. They feed the Comparisons tab.

CONTEXT FIELDS ARE DESCRIPTIVE

The interface states that these entries are written by the user and take no part in any calculation. They document the experiment for later review. Recording units and consistent names keeps experiments easy to compare. In this example the fields are still empty.

Data import and plate layout



What you see

The six-step import flow, file selectors with format options, a summary of imported data and a colour-coded plate map.

Main controls

Readings file, plate-map file, plate format, CSV separator, decimal symbol, Excel sheet and Validate and preview.

What it gives you

Each reading linked to a well, construct and condition. The example holds 450 readings from 18 wells and 9 constructs.

Data import: how to use it

The six steps

1. Choose a readings file and a plate-map file (CSV or XLSX).
2. Set the plate format (for example 96), the CSV separator and the decimal symbol. For Excel, name the sheet; if left empty, the first sheet is used.
3. Select Validate and preview. Validation only checks and previews the files; it saves nothing.
4. Review the correspondences: the plate view and the well table show the construct and condition each well received.
5. Correct any errors that are reported.
6. Confirm the import to attach the data to the experiment.

Expected inputs and what you see

Readings file

Columns Plate_ID, Well, Time_h, OD600 and OD_Blank. Fluorescence columns are optional: GFP, GFP_Blank and Autofluorescence_Estimate.

Plate-map file

Columns Plate_ID, Well and Construct_Name. Condition_ID is optional.

Plate view

Wells are coloured by construct. After analysis, marks show fit status: ✓ valid, × not valid, a dot not evaluated, C control. A thick outline marks the selection.

Well table

Lists each well with its construct, condition and number of readings (25 per well in the example).

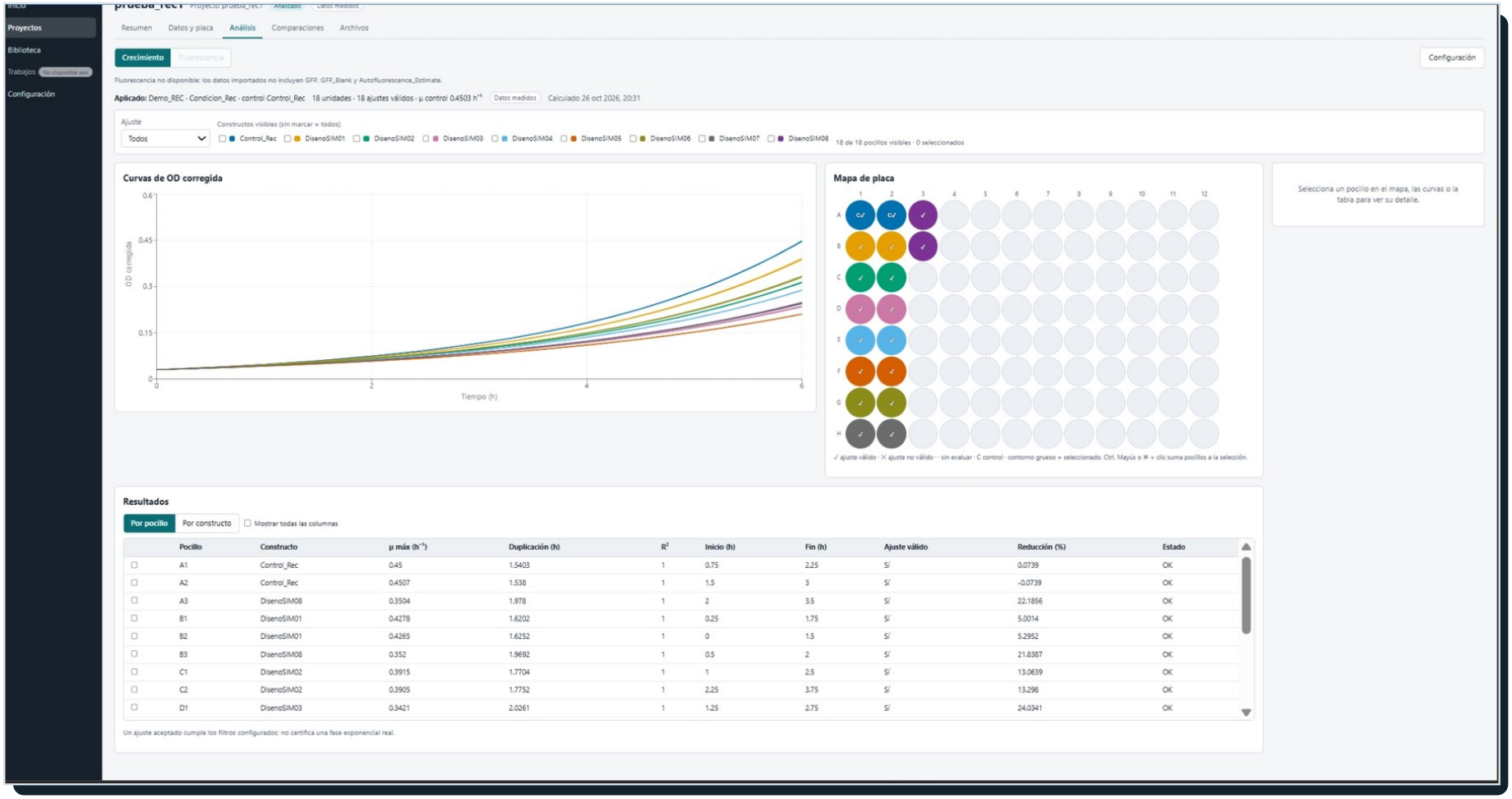
Sequence association

A panel for linking FASTA and GenBank sequences to constructs is reserved and marked as not yet available.

A CLEAN MATCH IS NOT A QUALITY CHECK

The example reports no discrepancies between readings and plate map. That confirms that the files line up, not that the experiment was well controlled. Blanks, time points and units still deserve a look in the source files.

Growth analysis



What you see

Growth/Fluorescence switch, a summary line, corrected-OD curves, a plate map with fit marks and a results table.

Main controls

Fit filter, construct checkboxes, well selection, by-well or by-construct table, all-columns option and Settings.

What it gives you

Growth-rate and doubling-time estimates per well and construct, with fit windows and reduction against the control.

Growth analysis: how to use it

How to use this view

1. Open Analysis (Análisis). Growth (Crecimiento) is the default view.
2. Read the summary line: plate, condition, control construct, number of units and valid fits, and the control growth rate (0.4503 h^{-1} in the example).
3. Narrow the view with the fit filter or the construct checkboxes. With nothing ticked, every construct is shown.
4. Select wells by clicking the plate map, a curve or a table row. Ctrl, Shift or ⌘ plus click adds wells to the selection.
5. Switch the table between By well and By construct, or show all columns.
6. Use Settings (Configuración) to review or adjust the analysis configuration.

Reading the results table

μ max (h^{-1})
Maximum specific growth rate estimated for the well.

Doubling (h)
Doubling time. In the example it equals ln 2 divided by μ: 0.45 h^{-1} gives 1.54 h.

R^2
Agreement between the fitted line and the data in the fit window. Values near 1 indicate a close fit.

Start / End (h)
The time window used for the fit. It can differ from well to well.

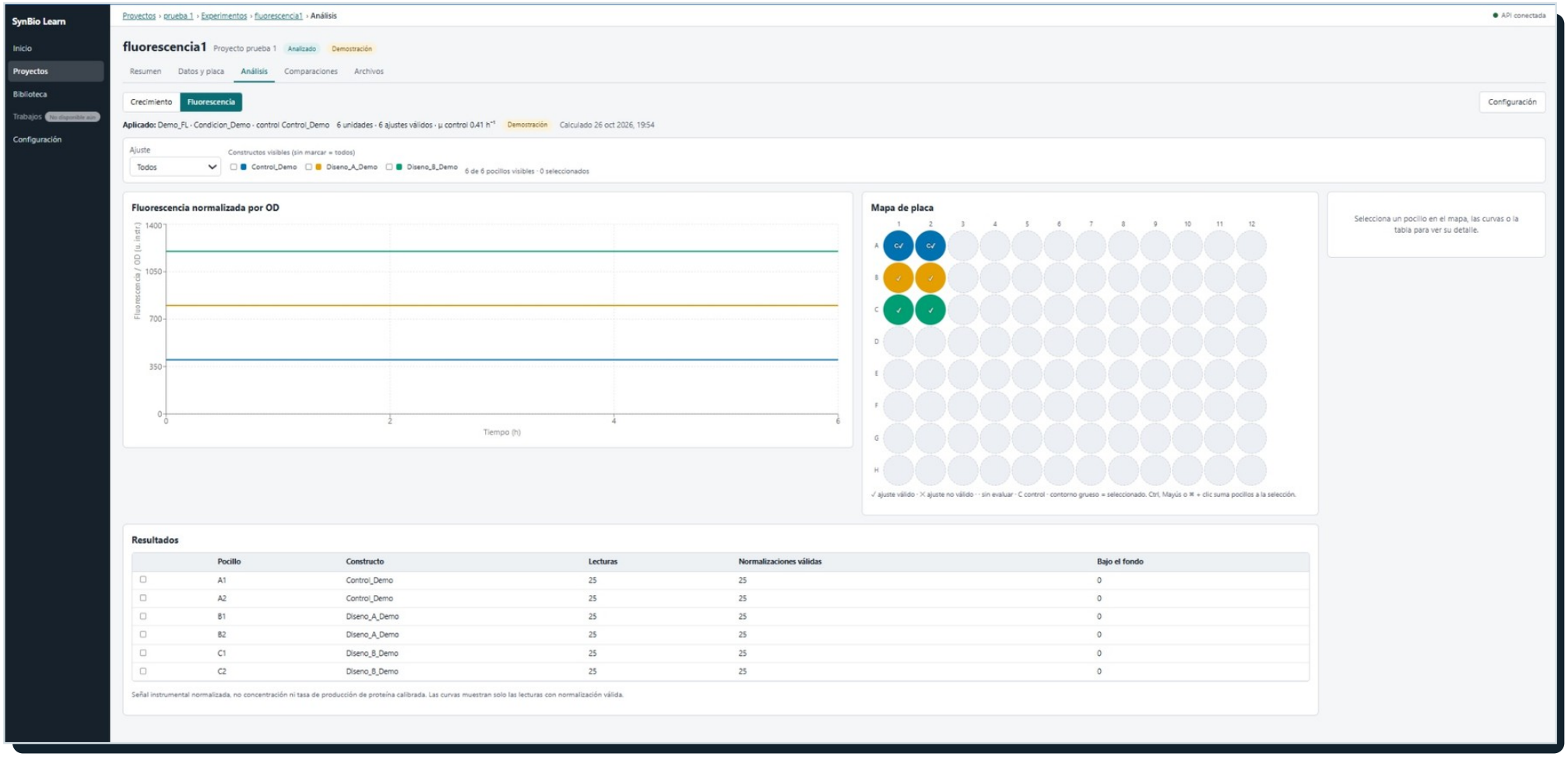
Valid fit
Whether the fit passed the configured acceptance filters.

Reduction (%)
How far μ falls below the control's. In the example, 0.3504 h^{-1} against a control of 0.4503 h^{-1} gives 22.2 %. Negative values mean faster growth than the control.

AN ACCEPTED FIT IS NOT PROOF OF EXPONENTIAL GROWTH

As the on-screen note says, an accepted fit meets the configured filters but does not certify a true exponential phase. Review the curve, the fit window and R^2 together before drawing conclusions.

Fluorescence analysis



What you see

Curves of fluorescence normalised by OD over time, the plate map and a table of valid normalisations.

Main controls

Growth/Fluorescence switch, construct filter and well selection.

What it gives you

Normalised instrumental signal per construct, with counts of readings and of values below background.

Fluorescence analysis: how to use it

How to use this view

1. Include fluorescence columns (GFP, GFP_Blank, Autofluorescence_Estimate) in the readings file. Without them the Fluorescence tab stays disabled and the screen names the missing columns.
2. Open Analysis and choose Fluorescence (Fluorescencia).
3. Pick constructs or wells with the filters and the plate map, as in the growth view.
4. Read the curves, then check the table for valid normalisations and values below background.

Reading the results

Fluorescence / OD
Fluorescence signal divided by optical density, in instrument units. It is a normalised reading, not a concentration.

Readings
Number of readings recorded for each well (25 in the example).

Valid normalisations
Readings for which the normalisation could be computed. The curves display only these.

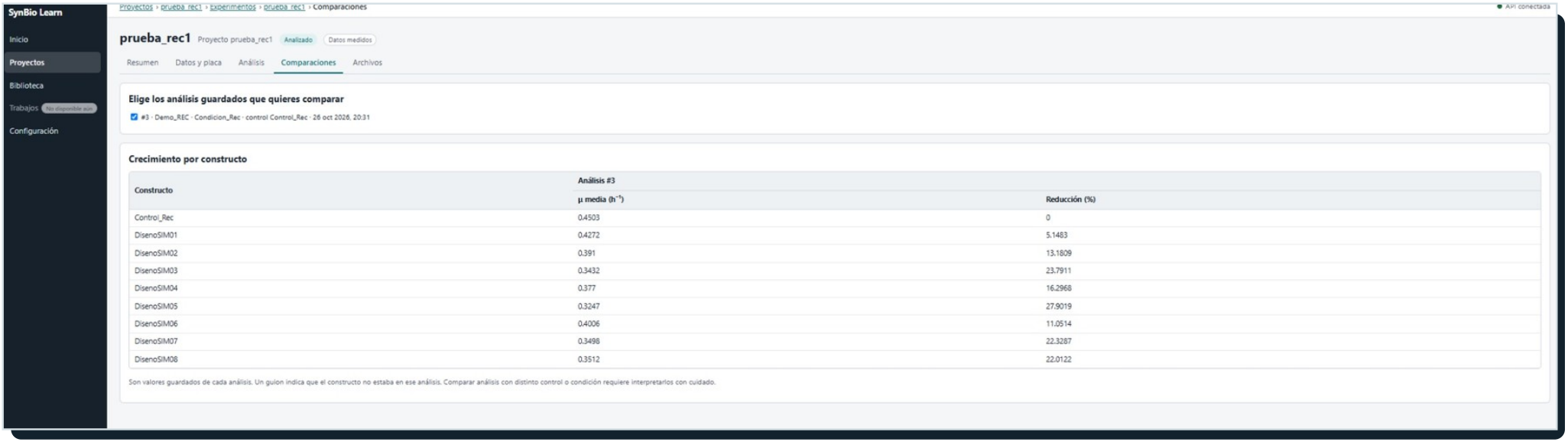
Below background
Readings that fall below the background level. The example shows 0.

Summary line
Names the plate, condition and control, as in the growth view.

NORMALISED SIGNAL, NOT PROTEIN CONCENTRATION

The signal is instrumental. It is neither a calibrated concentration nor a protein-production rate. The example is a demonstration dataset (badge: Demostración) and its constant traces illustrate the layout, not a result.

Saved comparisons



What you see

A list of saved analyses to tick, and a table of growth by construct.

Main controls

Analysis checkboxes. Each analysis is identified by number, plate, condition, control and date.

What it gives you

Mean growth rate and reduction against the control for each construct, side by side across analyses.

Saved comparisons: how to use them

How to use this view

1. Open Comparisons (Comparaciones).
2. Tick the saved analyses to include. Each entry shows its number, plate, condition, control and date, for example #3 · Demo_REC · Condicion_Rec · control Control_Rec.
3. Read the Growth by construct table: each analysis has a column for mean growth rate and one for reduction.
4. A dash means that the construct was not part of that analysis.

Reading the table

Saved analysis
A stored set of results, identified by number, plate, condition, control and date.

μ mean (h^{-1})
Mean growth rate of the construct in that analysis.

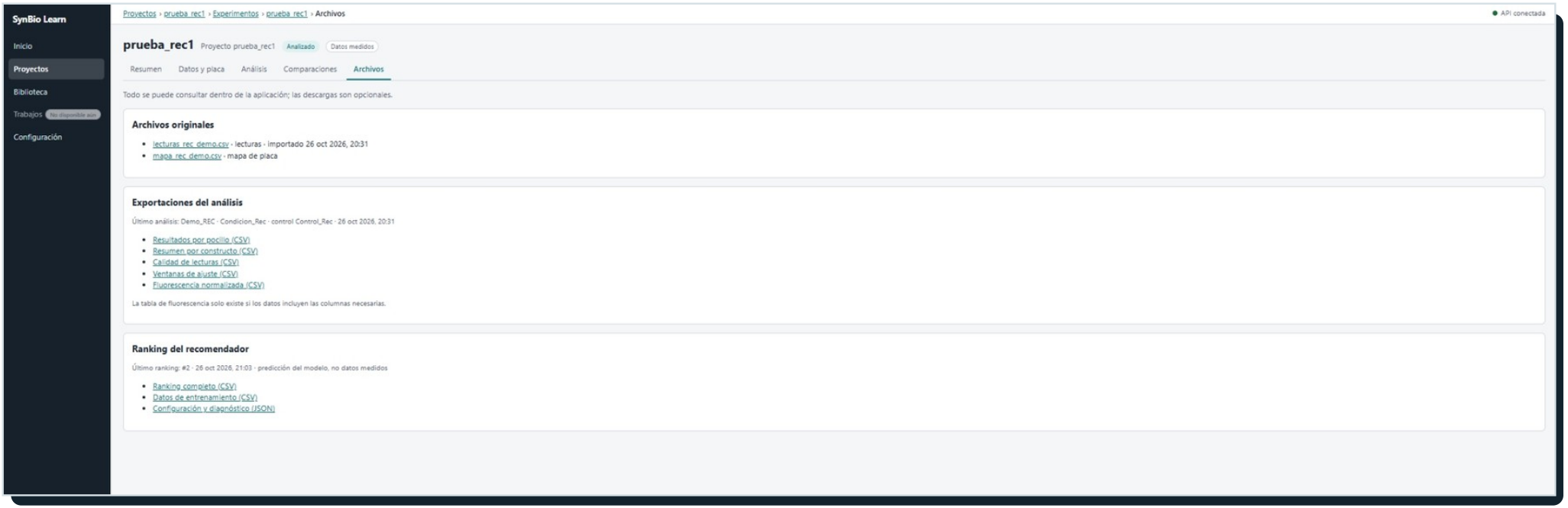
Reduction (%)
Decrease in growth rate relative to the control of the same analysis. The control itself shows 0.

Example reading
The control grows at 0.4503 h^{-1}. The eight designs show reductions from 5.1 % (DisenoSIM01) to 27.9 % (DisenoSIM05).

COMPARE LIKE WITH LIKE

Comparing analyses that use a different control or condition requires care, as the screen itself reminds you. The view lists saved values only and does not include statistical tests.

Files and exports



What you see

Three groups: original files, analysis exports and recommender ranking files.

Main controls

Links to open each file. Downloads are optional because everything can be consulted in the app.

What it gives you

Traceable inputs, CSV outputs and, for the recommender, ranking, training-data and diagnostics files.

Files and exports: how to use them

How to use this view

1. Open Files (Archivos).
2. Under Original files, find the readings and plate-map files imported into the experiment.
3. Under Analysis exports, choose among the outputs of the latest analysis. Its plate, condition, control and date are shown.
4. Under Recommender ranking, find the files of the latest ranking (#2 in the example).
5. Everything can be consulted inside the application; downloads are optional.

What each export contains

Results by well (CSV)
The per-well results of the latest analysis.

Summary by construct (CSV)
The construct-level summary.

Reading quality (CSV)
Information on the quality of the readings.

Fit windows (CSV)
The time windows used for the fits.

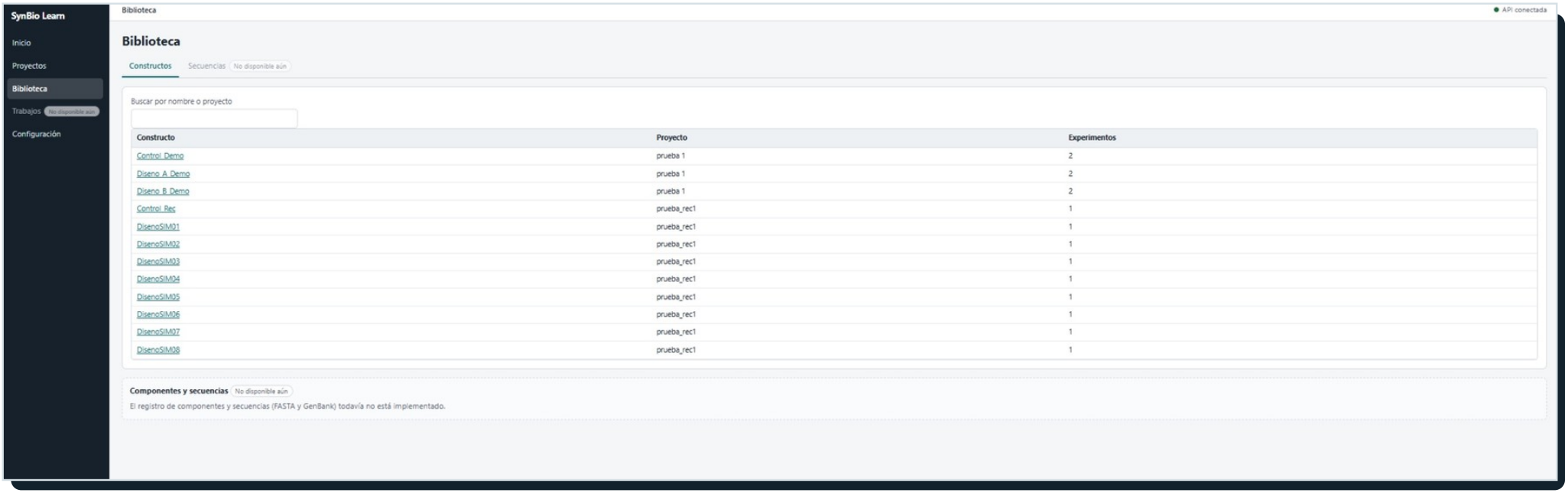
Normalised fluorescence (CSV)
Available only when the data include the required fluorescence columns.

Recommender files
Full ranking (CSV), training data (CSV) and configuration and diagnostics (JSON), labelled as model prediction, not measured data.

PREDICTIONS ARE LABELLED AS PREDICTIONS

The interface marks recommender output as model prediction rather than measured data. The machine-learning methods behind the ranking are still in development (see the roadmap), so the ranking is a development-stage feature.

Construct library



What you see

A Constructs tab with a search field and a table of constructs, projects and experiment counts. A Sequences tab is marked as not yet available.

Main controls

Search by name or project. Construct names are clickable.

What it gives you

A cross-project overview of every construct and where it has been used.

Construct library: how to use it

How to use this view

1. Open Library (Biblioteca).
2. Stay on the Constructs tab to list every construct recorded.
3. Type in the search field to filter by construct name or project.
4. Read the Experiments column to see in how many experiments each construct appears.

What the table shows

Construct
A genetic design or strain variant assigned to wells through a plate map.

Project
The project in which the construct was recorded.

Experiments
The number of experiments in which the construct appears. In the example, the three Demo constructs appear in two experiments each; the control and eight designs of prueba_rec1 in one each.

Sequences tab and registry
Marked as not yet available. Components and sequences (FASTA, GenBank) are planned with the sequence module.

BUILT FOR TRACING CONSTRUCTS ACROSS EXPERIMENTS

The library gathers constructs across projects, so a design can be followed from one experiment to the next. Sequence-level information will be attached once the sequence module is integrated.

Glossary of terms and output columns

Construct
A genetic design or strain variant assigned to wells through a plate map.

Control
The reference construct against which growth is compared.

Plate map
A table linking each well to a construct and, optionally, a condition.

Well
One position of the microplate, named by row and column, for example A1.

OD600
Optical density at 600 nm, a standard indicator of culture density.

OD blank
A reading of medium without cells, used to correct the OD600 for background.

Corrected OD
The OD600 corrected for the blank; the quantity plotted in the growth curves.

Data-origin badge
A label beside the experiment name showing whether the data are, for example, measured or a demonstration.

Specific growth rate, μ
Rate of increase of biomass per hour (h^{-1}) during exponential growth.

Doubling time
Time needed for the culture to double: ln 2 divided by μ.

R^2
Agreement between the fitted line and the data in the fit window; closer to 1 means a closer fit.

Fit window
The time interval of the curve used to estimate μ, given as Start and End (h).

Reduction (%)
Fall in μ relative to the control. Negative values indicate faster growth than the control.

Normalised fluorescence
Fluorescence divided by OD, in instrument units. It is not a concentration.

Saved analysis
A stored set of results identified by number, plate, condition, control and date.

ABOUT THESE DEFINITIONS

These are general descriptions of the terms as they are used in the interface. They are not a specification of the application’s algorithms.

Where the project is heading

Current prototype — IN PROTOTYPE → **Sequence module** — IN PROGRESS → **Jobs system** — PLANNED

Machine learning · Biophysical models — IN DEVELOPMENT

Sequence viewer

IN PROGRESS

A viewer to inspect the sequences associated with experimental constructs. The interface already reserves places for it: a Sequences tab in the library and a panel for linking FASTA and GenBank sequences to constructs in the import flow. Supported formats and viewer functions will be documented once the module is integrated.

Machine learning

IN DEVELOPMENT

Methods that learn from accumulated experimental results to support predictive analysis and the prioritisation of candidate constructs. A recommender section already exists: it lists ranking, training-data and configuration and diagnostics files, always labelled as model prediction. Methods, features and evaluation will be documented as they mature.

Biophysical models

IN DEVELOPMENT

Biophysical modelling is a further analytical layer under development. Its biological scope, assumptions and parameters will be described together with the implemented models, so that their outputs can be interpreted correctly.

Jobs

PLANNED

A monitor for long calculations running in the background, planned after the sequence module. It is described on the next two pages.

Jobs: monitoring long calculations

DESIGN STAGE · NOT AVAILABLE YET

The Jobs section (Trabajos in the interface) is reserved for supervising long calculations that the application runs in the background. It is a list of calculation tasks, not of projects or experiments. Today it appears as “not yet available” because the system has not been developed.

For example, when a user requests descriptors for many sequences or the training of a model, the idea is that the application stays usable while the calculation continues.

Illustrative tasks (planned design, not current functions)

TASK	CONTEXT	STATUS	PLANNED ACTION
Compute sequence descriptors	Project A · selected set	Running	View progress or request cancellation
Train a model	Project A · training data	Pending	Inspect the configuration
Analyse a plate	Project B · experiment 3	Completed	Open the results
Compute a ranking	Project A · analysis 5	Failed	Inspect the error

How Jobs differs from the other sections

Projects

Organises experiments and results.

Learning (planned section)

Where the training or the ranking would be configured and requested.

Jobs

Follows execution: what is pending, running, completed or failed.

Jobs: design goals and boundaries

Calculations would be requested from their own screen, and their status could be consulted there or in the global list. Jobs would not be needed to configure each calculation. The design aims for the following behaviour.

1. **Keep working**
   Changing tab should not interrupt a calculation.

2. **No repeated work**
   Consulting a result again should display it, not run the calculation a second time.

3. **Recoverable failures**
   A failure should leave a message that can be consulted afterwards.

4. **Closing the page**
   Closing the browser should not cancel a job.

RECOVERY BOUNDARIES

Closing the browser is different from stopping the server. A job should survive the closing of the page, but continuing after the computer has been switched off will require a specific recovery mechanism. That mechanism is part of the design work still to be done.

What works today

Analyses and the recommender keep their current run buttons. The Jobs system will be developed after the sequence module, following the order agreed for the project.

Quality, scope and availability

Ongoing quality work

- Documenting data provenance, analysis settings, units and calculation methods, including fit-acceptance filters and normalisation rules.
- Checking calculations against independently computed reference examples.
- Testing how the application behaves with missing, invalid or unexpected input files.
- Evaluating future predictive methods against appropriate reference methods before reporting their performance.
- Keeping this guide aligned with the implemented features as new modules are integrated.

Availability

The application source code is maintained privately and is not distributed with this guide. A demonstration and selected results are available on request through the project owner. Access to the source code, if any, would be considered separately.

SynBio Learn is a development-stage prototype. This document does not grant a licence to use, modify or redistribute the software.

SCOPE OF THIS GUIDE

This guide documents the interface and workflow of the prototype. The screenshots illustrate example datasets and are not experimental results. No claim is made about predictive performance, about compatibility with every instrument export, or about the validation of the calculations beyond what is described here.